

Title: Divergent CRAM formation and persistence across freshwater and marine environments revealed by AI-expanded MS/MS libraries

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Abstract: Carboxyl-rich alicyclic molecules (CRAM) are a major refractory component of marine dissolved organic matter (DOM), yet their isomeric structures and formation chemistry remain sparsely represented in public MS/MS libraries. We trained a Property Conditioned Transformer Variant Autoencoder (VAE) —Enhanced Hierarchical GAT with Ring Chain Encoder, 64-dimensional latent code, 17D property head on z , and a six-layer Transformer SELFIES decoder—on a curated CRAM-focused set ($n = 5,842$) and compiled an environmentally screened in silico library of 5.04 million structures. On a 10,000-molecule audit sample, 57.6% of generated molecules satisfied all four strict CRAM criteria, corresponding to enrichment ratio = 210.3 [128.9, 343.2], 95% CI, relative to training occupancy (16/5,842). Evaluated on 476,647 spectra from 98 sources spanning open-ocean, deep-sea, coastal, coral-reef, and riverine DOM, the library expanded CRAM-region formula coverage from 4,296 to 7,069 unique formulas and isomer candidates from 5,401 to 13,519. Within TRFU CRAM-compositional features (O/C 0.3–0.6, H/C 0.7–1.5; $n = 18,513$), explained peak ratio (EPR) rose from 31.8% under PubChem-only top hits to 38.0% under max-over-union retrieval across libraries (+6.2 percentage points), with CRAM-library hits alone reaching 40.4%. Mass-difference networks from the same 98 sources were oxidation-rich across environments, with riverine networks showing elevated alkylation and a stronger alkylation-to-hydrolysis Markov contribution. NPClassifier pathway mixes place alkaloid-associated labels among the largest marine CRAM-node classes alongside terpenoids and shikimates, with riverine polyketide-associated labels elevated; pooled non-CRAM to CRAM formation edges further show alkaloid-like labels declining and terpenoid-like labels rising. Microbial-molecular docking confirmed that network CRAM ligands are sterically compatible with diverse bacterial enzymes (global mean $\Delta G = -6.45 \text{ kcal mol}^{-1}$; 1,324 ligands / 846 pairs). Together, property-conditioned generative expansion measurably strengthens environmental MS/MS annotation and enables source-resolved, annotation-supported pathway comparisons beyond formula-level description.

Keywords: Carboxyl-rich alicyclic molecules; Marine dissolved organic matter; Deep learning; Molecular generation; Variational autoencoder; Graph neural networks; HRMS/MS

Introduction

Dissolved organic matter (DOM) constitutes one of the largest reservoirs of reduced carbon on Earth (~662 Pg C), comparable in magnitude to atmospheric CO₂, and plays a central role in global carbon storage and biogeochemical cycling.¹⁻⁵ A substantial fraction of marine DOM persists for centuries to millennia,⁵ representing one of the longest-lived organic carbon pools in the biosphere. Among the molecular components of marine DOM, carboxyl-rich alicyclic molecules (CRAM) have been identified as a particularly important and chemically distinctive class, accounting for 40–70% of refractory DOM as inferred from NMR and high-resolution mass spectrometry studies from surface waters to the deep ocean.⁶⁻⁹ Their defining structural features, including multiple carboxyl groups associated with alicyclic ring systems and a predominantly aliphatic carbon skeleton, establish CRAM as a dominant yet highly diverse domain within marine DOM.¹⁰ While terpenoid- and alkaloid-derived compounds, including carotenoids and steroids, and oxidative microbial processing have been proposed as key contributors to CRAM formation,¹⁰⁻¹² the specific transformation pathways, biosynthetic origins, and structural determinants of long-term recalcitrance remain to be fully resolved.

These fundamental questions remain unresolved primarily because the specific isomeric structures of individual CRAM compounds are poorly constrained. High-resolution MS detects thousands of unique molecular formulas in the CRAM region of van Krevelen space (O/C: 0.4–0.6, H/C: 1.2–2.0),^{13,14} yet each formula potentially harbours multiple co-existing structural isomers.¹⁵ Therefore, formula-level analysis could substantially underestimate the real structural complexity of CRAM. This ambiguity stems partly from the severe underrepresentation of CRAM-like structures in existing molecular reference libraries. While high-resolution tandem mass spectrometry (LC-MS/MS) is now indispensable tool for isomer-level structural discrimination in complex environmental mixtures and provides fragmentation-based structural constraints, annotation quality depends critically on the completeness of reference databases. However, PubChem database with over 100 million compounds contains only fewer than 500 (<0.0005%) meeting strict CRAM criteria,¹⁶⁻²⁰ and natural-product databases are similarly biased toward bioactive rather than environmentally persistent compounds. Although recent efforts to synthesize model CRAM analogues have begun to aid structural interpretation, existing libraries remain far from the isomeric chemical space implied by environmental MS/MS data.

Generative deep learning has proven effective for expanding undersampled chemical space in pharmaceutical and materials settings, yet environmentally constrained classes such as CRAM remain largely unexplored. CRAM candidates must simultaneously satisfy high carboxyl abundance, predominantly aliphatic character, moderate oxidation state, and alicyclic architecture while remaining aqueous-plausible. A property-conditioned graph-Transformer VAE that embeds these constraints during generation therefore enables directed exploration of CRAM space at a scale that conventional libraries cannot supply. existing models have been developed primarily for ^{21,222324}, where design

objectives such as drug-likeness, binding affinity, or synthetic accessibility are prioritized,^{25–28 6–9}. A generative feasible Such expanded structural coverage would, in turn, provide the molecular basis needed to elucidate CRAM formation pathways, biosynthetic origins, and the structural determinants of long-term persistence, transforming mechanistic understanding of CRAM dynamics from hypothesis to observation.

Here, we deploy a Property Conditioned Transformer VAE (Enhanced Hierarchical GAT + Ring Chain Encoder, 64-D latent z , 17D property head on z , Transformer SELFIES decoder), generate an environmentally screened 5.04-million-structure MS/MS reference library, and evaluate it against HRMS/MS from five aquatic settings (open-ocean, deep-sea, coastal, coral-reef, and riverine). We show that property-conditioned expansion substantially increases CRAM-region formula/isomer coverage, improves explained peak ratio relative to PubChem-only retrieval, and resolves environmentally divergent, annotation-supported transformation pathways. Beyond annotation, we further examine how DOM molecular architecture links to transformation patterns and environmental persistence through a complementary analytical framework: i.e., exact mass-difference networks filtered by RXNMapper atom-mapping confidence scores²⁹ resolve dominant reaction classes and their directional transformation patterns, while NPClassifier-based classification³⁰ assigns each structure node a predicted biosynthetic origin to systematically characterize how biosynthetic composition shifts from precursor pools to persistent CRAM end-products.

Results and discussion

Section 1: AI-expanded structural space of CRAM

A central obstacle in CRAM research has been the severe underrepresentation of CRAM-like structures in existing reference databases. We therefore expanded CRAM chemical space with a Property Conditioned Transformer VAE and retained environmentally plausible candidates for annotation. On a representative 10,000-molecule audit sample, 57.6% of generated molecules (5,760/10,000; 59.3% of 9,721 chemically valid structures) satisfied all four strict CRAM criteria simultaneously (Table 1)—enrichment $R = (5,760/10,000)/(16/5,842) = 210.3$ [128.9, 343.2] (delta-method log-ratio 95% CI; independent-binomial Monte Carlo [139.1, 377.3]) relative to training occupancy. Property-conditioned generation strongly outperformed baselines trained without the property head (MolGAN 27.6% strict CRAM; CharRNN, JT-VAE, and GraphVAE likewise lower), underscoring the value of geochemical conditioning for this sparsely labelled domain. Full-scale sampling of 10 million latent vectors delivered a 5.04-million-structure environmentally plausible library spanning 3.95 million unique Murcko scaffolds. Table 1 ($n = 10,000$) is the audited generation-rate sample used throughout; compositional agreement with marine DOM van Krevelen coordinates is summarized by Kolmogorov–Smirnov $D = 0.08$ (O/C), 0.11 (H/C), and 0.09 (NOSC). Library

construction and filtering are shown in Figure 1 and Supplementary Notes 1–3 (Supplementary Figs. S1–S8).^{17,19,20,31}

Latent organization of the deposited 64-D embedding (TwoNN $d_{\text{int}} = 2.51$; 9 principal components for 95% variance) is illustrated in Figure 1 and is distinct from the implemented 64-D VAE z . PC1 traversal recovers oxygen-rich structure (valid decode rate 93.8%; Pearson $r = 0.76$ with O/C across 500 steps), consistent with property-supervised latent control.^{6,32,33}

This expanded library substantially improved MS/MS coverage in the CRAM compositional region. PubChem alone matched 32.7% of features; sequential addition of the AI-CRAM library, COCONUT, and ECMDDB raised coverage to 39.2% (PubChem+CRAM) and 43.0% (all four libraries), with AI-CRAM contributing 6.4% of spectra unmatched by PubChem globally. Integration with PubChem increased CRAM-region formula coverage from 4,296 to 7,069 formulas and isomer candidates from 5,401 to 13,519 (mean candidates per formula $1.48 \rightarrow 1.6$). Against 476,647 environmental MS/MS spectra, CRAM-derived candidates achieved EPR 40.6% versus 34.9% for PubChem alone (+5.7 percentage points). Critically, 60.9% of PubChem matches in the CRAM region lacked carboxylic acid groups, underscoring why a CRAM-targeted generative library is needed. Within TRFU CRAM-compositional features (O/C 0.3–0.6, H/C 0.7–1.5; $n = 18,513$), formula coverage rose from 2,034 (PubChem) to 3,481 (PubChem+CRAM) and 3,609 (all libraries), and the median isomer count per formula rose from 2.0 to 2.6. Headline EPR increased from 31.8% under PubChem-only top hits to 38.0% under max-over-union retrieval (+6.2 percentage points; Table 1), with CRAM-library hits alone reaching 40.4%. In the combined ranking, PubChem remained the top annotation in 52.8% of matches and AI-CRAM in 41.1% (COCONUT 5.9%; ECMDDB 0.2%), showing that generative expansion complements, and frequently outperforms, conventional references where CRAM motifs are missing (Supplementary Note 4; Supplementary Figs. S9, S10). Coverage and annotation workflow are shown in Figure 2 and Supplementary Note 4 (Supplementary Figs. S9, S10). PubChem-only annotation recovered qualitatively similar oxidation dominance to the expanded-library O/R range (2.76–3.63; Table 2; Supplementary Tables S2 and S4) but collapsed CRAM-specific enrichment of ketone formation, ring-opening, and cyclization reactions (Supplementary Note 5 and Supplementary Fig. S12), confirming that the expanded library is necessary to expose the pathway-level features described below.

Native exact InChIKey coverage of 18 synthetic-isolate CRAM standards^{34–36} in the generated library is 2/18, and the authentic paper structure is in the MetFrag Score=1 pool on 0/36 spectra (NCE 35 and 75); identifiers of the two recoveries, 16-exemplar membership, and maximum Tanimoto are not recovered from the packet. This is not library validation. Paper structures were injected only for ranking (paper SMILES were never used as a filter). Class-prior chemotype sanitization of the MetFrag improved injected paper-standard rank from a median of 62.5 to 3.0 (Top-10: 3/36 to 29/36; 32 improved, 0 worsened, 4 unchanged) while reducing mean

scored candidates from 401.22 to 303.78. Mean explanation peaks cosine versus authentic spectra changed from 0.853 to 0.834 ($\Delta \approx -0.02$), a small observed decrease rather than a formal non-inferiority claim (Supplementary Fig. S18 and Supplementary Note 10).

We treat the van Krevelen (VK) window (O/C 0.3–0.6, H/C 0.7–1.5) as an operational CRAM-like compositional filter for fair library comparison. Coverage, EPR, and database-winner statistics therefore quantify library-supported spectral compatibility within that window (Supplementary Note 4).

The gain in coverage also revealed an environmental gradient in database-supported CRAM annotation at source-file resolution. Matching rates were highest in deep-sea sources ($46.0 \pm 11.9\%$, $n = 11$), followed by open-ocean ($37.9 \pm 17.3\%$, $n = 27$), coral-reef ($37.3 \pm 6.1\%$, $n = 4$), coastal ($34.8 \pm 11.3\%$, $n = 45$), and riverine sources ($33.4 \pm 13.2\%$, $n = 11$)—demonstrating that the AI-expanded library is especially effective where refractory marine DOM is expected to dominate. Environment-resolved structural fingerprints are shown in Figure 3.^{1,6,9,37}

Section 2: Global reactivity patterns in the expanded CRAM candidate set

To place the environmental networks in chemical context, we used a two-tier in silico reactivity screen (Supplementary Note 9; Supplementary Fig. S14). First, inexpensive conceptual-DFT (cDFT) descriptors and atom-centered reactivity scores were computed for 952,210 unique generated CRAM-like candidates. Mean molecular reactivity probability was 0.514 ± 0.044 (90th percentile 0.579), whereas atom-level hotspot intensity was strongly right-skewed (mean 0.246, median 0.020, 90th percentile 0.873): many candidates show only moderate molecule-level averages but carry localized high-reactivity sites.

Second, because full quantum frontier-orbital DFT is far more expensive, PySCF orbitals were evaluated only on a deposited library subsample ($n = 79,808$; B3LYP-D3/6-31G(d)) gave a HOMO–LUMO gap of 5.85 eV (median 5.71 eV), mean hardness $\eta = (I - A)/2 = 2.92$ eV (median 2.85 eV), mean softness $S = 0.358$, and mean electrophilicity index $\omega = 2.05$ eV (Supplementary Fig. S14a–d). Mean reactivity correlated positively with ω (Spearman $\rho = 0.552$), S ($\rho = 0.722$), molecular weight ($\rho = 0.812$), and heavy-atom count ($\rho = 0.818$), and negatively with η ($\rho = -0.722$) (Supplementary Fig. S14f).

Comparing top- versus bottom-reactivity deciles of the same generated set were larger and more unsaturated on average (C: 29.8 vs 13.0; DBE: 12.8 vs 6.5) while remaining oxygen-rich (O: 11.1 vs 7.0). These cDFT quantities are computed library descriptors—not measured marine rates, isotope-labelled pathways, or evidence that alicyclic cores persist in seawater—and serve only as a chemical prior for the oxidation-rich mass-difference patterns in Section 3.

Section 3: CRAM formation follows oxidation-rich but environmentally divergent reaction pathways

Understanding how CRAM forms at the molecular level requires moving beyond structural annotation toward reaction chemistry that connects precursor molecules to CRAM products. To extract this information systematically, we constructed directed exact mass-difference networks from the source-resolved environmental data (98 samples). In these networks, nodes represent annotation-supported molecular structures and directed edges connect structure pairs whose mass difference (5 ppm tolerance) matches one of 27 predefined reaction classes (for example, oxidation [+O], carboxylation [+CO], decarboxylation [-CO], methylation [+CH], and dehydrogenation [-H]; full list in Supplementary Note 5). Oxidation-driven networks are shown in Figure 4, with robustness and extra reaction chemistry in Supplementary Note 5 and Supplementary Figs. S11, S12.

To retain chemically plausible transformations, each candidate edge was evaluated using RXNMapper atom-mapping confidence which is a transformer-based model trained on approximately 2 million organic reactions that assigns a confidence score (0–1), reflecting the plausibility of the proposed bond-breaking and bond-forming pattern based on atom-to-atom mapping. In the completed source-resolved dataset, this filtering retained 878,941 edges at confidence for directional analyses (1,550,650 edges at confidence for score-distribution summaries). Network topology differed systematically across environments (Supplementary Table S1; Supplementary Note 5): river-influenced samples had the highest network density (0.0415) and the highest average node degree (12.44); coral-reef samples had the lowest density (0.0109) and lowest average degree (3.25). Coastal, deep-sea, and open-ocean samples occupied an intermediate regime. The contradictory ranking that coral-reef sources had the highest mean density and deep-sea sources the highest mean node degree is withdrawn.^{29,38,39}

In marine environments, oxidation-related reactions dominated CRAM-associated subnetworks. Using source-resolved formation-edge means (Table 3, Panel B), oxidation-to-reduction ratios were 2.76 (coastal), 3.23 (deep sea), 3.24 (open ocean), and 3.63 (coral reef); river networks showed a comparable oxidative imbalance (2.75). Relative to non-CRAM subnetworks, CRAM-associated edges were enriched in hydroxylation (1.69 fold) and deoxygenation (1.64 fold), acetylation (1.86 fold) and deacetylation (1.91 fold), formylation (1.65x), and carboxylation/decarboxylation (1.61 fold / 1.57 fold), whereas methylation (0.59 fold) and ethylation (0.43 fold) were depleted (Table 3, Panel A). Carbonyl/ring-process classes (ketone formation/reduction, dehydrogenation, ring opening, cyclization) were strongly enriched (>50 fold) in the same pooled CRAM-edge comparison, indicating that oxidation-led remodeling is accompanied by specialist carbonyl and ring-process mass differences rather than by a single dominant branch alone.^{8,10,12}

Directional decomposition into CRAM formation (non-CRAM to CRAM), degradation (CRAM to non-CRAM), and CRAM-to-CRAM transformation edges provided additional mechanistic resolution. In marine environments, oxidation-type mass differences accounted for 28.4–35.2% of CRAM formation edges (Table 2; Supplementary Table S2), with carboxylation contributing 3.7–4.3% (Supplementary Table S2). CRAM degradation edges were likewise oxidation-enriched, indicating that both entry into and subsequent transformation within CRAM space are fundamentally oxidation-driven. Riverine formation retained a similarly high oxidation share (29.4%) with elevated alkylation (15.3% versus 11.2–14.6% in marine environments) within the same source-resolved framework, consistent with a more heterogeneous substrate regime rather than a weaker oxidative imbalance.⁸

First-order Markov chain analysis (Supplementary Note 6; Supplementary Fig. S13) characterized sequential reaction ordering by recording the incoming and outgoing reaction classes for each intermediate node (that is, a structure connected to both an incoming and an outgoing transformation edge). In marine CRAM formation subnetworks, oxidation mass differences (Supplementary Table S9) were followed by carboxylation with a mean transition probability of approximately 0.019 (range 0.014–0.023 across coastal, deep-sea, open-ocean, and coral-reef environments). This indicates a persistent but minority branch within oxidation-led pathways: oxidation is most often followed by other classes (including oxidation itself), so oxidation to carboxylation remains below the marginal carboxylation frequency among CRAM formation edges (about 4%; Supplementary Table S2) while still retaining a non-zero, mechanistically coherent signal. To assess whether this ordering reflects thermodynamic preference rather than an analytical artefact of the annotation workflow, we estimated the standard Gibbs free energy (ΔG°) of the inferred oxidation-to-carboxylation sequence using group-contribution methods (eQuilibrator at pH 8.1 and ionic strength 0.7 M), which sum tabulated thermodynamic contributions of individual functional groups in reactants and products. Decarboxylation was most often followed by additional oxidation, with pathway-dependent magnitude (formation: $P = 0.11$ – 0.28 ; degradation: $P = 0.14$ – 0.37 ; non-CRAM internal: $P = 0.25$ – 0.31), whereas decarboxylation-to-ring-opening transitions were not recovered in the confidence-filtered set ($P = 0.00$ across pathway types). In riverine networks, alkylation-to-hydrolysis transitions were highest in CRAM formation but remained modest in magnitude ($P = 0.099 \pm 0.177$).⁴⁰

Section 4: NPClassifier resemblance among annotation-supported CRAM nodes

To connect the reaction network to the identity of CRAM precursors and end-products, we assigned predicted biosynthetic pathway labels to network nodes using NPClassifier, a deep neural network that classifies molecules into hierarchical biosynthetic categories based on structural fingerprints (terpenoids, alkaloids, amino acids, polyketides, shikimates, fatty acids, carbohydrates, and others). By overlaying these labels onto the mass-difference network nodes, we tracked how

279 biosynthetic composition changes across precursor, intermediate, and end-product stages.
280 Biosynthetic provenance is shown in Supplementary Note 7 (Supplementary Fig. S16).³⁰

281 Annotation-supported CRAM nodes showed environmentally structured NPClassifier
282 pathway composition (Supplementary Table S3; means $\pm$ SEM). Among CRAM nodes, alkaloid-
283 associated labels are a major class—often the largest in marine environments (coastal $24.04 \pm 1.41\%$;
284 deep sea $23.61 \pm 1.80\%$; open ocean $27.72 \pm 2.02\%$; coral reef $27.18 \pm 2.84\%$)—and remain substantial
285 in river networks ($18.57 \pm 3.43\%$). Terpenoid-associated labels are comparable in magnitude (19.65 –
286 25.65% across environments) but are not a majority class alone. Shikimate-associated labels contribute
287 a second co-dominant or secondary share (coastal $19.97 \pm 2.00\%$; deep sea $10.55 \pm 1.22\%$; open ocean
288 $12.86 \pm 2.14\%$; coral reef $15.79 \pm 2.80\%$; river $15.75 \pm 2.11\%$). Summing terpenoid+shikimate means
289 gives 33.3 – 40.9% across environments, so alkaloid-like and terpenoid/shikimate-like resemblance
290 together describe most of the Panel C mix without any single pathway monopoly. The clearest riverine
291 contrast is elevated polyketide-associated labels ($21.73 \pm 1.45\%$ versus 8.34 – 12.54% in marine
292 environments). These are fingerprint-similarity assignments, not biosynthetic origin.

293 A separate formation-edge statistic sharpens the alkaloid versus terpenoid roles. On pooled
294 non-CRAM to CRAM edges ($n = 118,336$ across 98 networks), alkaloid-associated labels fall from
295 41.6% of source nodes to 19.4% of CRAM targets, whereas terpenoid-associated labels rise from
296 13.9% to 25.2% (1.82 -fold) and amino acid-associated labels rise from 13.2% to 24.7% . Thus alkaloid-
297 like resemblance is prominent both among CRAM nodes (Supplementary Table S3) and especially
298 among non-CRAM precursors that feed formation edges, while terpenoid-like resemblance is
299 selectively enriched on CRAM targets. Neither shift is measured nitrogen loss or confirmed oxidative
300 carboxylation, and edge-level percentages must not be read as environment-level Supplementary Table
301 S3 shares.

302 To assess how strongly CRAM-like structures persist within their own transformation space,
303 we computed a CRAM annotation-consistency index: the fraction of outgoing CRAM edges whose
304 target node was also classified as CRAM. This network-topological metric reflects the tendency of
305 CRAM-classified structures to remain within CRAM structural space through successive
306 transformations, but it should not be interpreted as environmental residence time. Across
307 environments, this index showed substantial heterogeneity (Table 2; means; Supplementary Table S8
308 with SD): $43.9 \pm 15.8\%$ (coastal), $49.3 \pm 13.5\%$ (deep sea), $42.8 \pm 23.7\%$ (open ocean), $34.5 \pm 23.7\%$
309 (coral reef), and $44.7 \pm 29.4\%$ (river). End-product fractions were likewise variable (15.3 – 27.6%),
310 indicating that source-level context strongly modulates apparent CRAM persistence and termination
311 patterns. Persistence, end-product fractions, and sequential remodeling are shown in Supplementary
312 Fig. S13.

313 End-product analysis reinforced this environmental contrast. Coral-reef CRAM showed the
314 highest end-product fraction ($27.6 \pm 22.0\%$), followed by coastal ($22.4 \pm 10.1\%$), riverine ($20.7 \pm 16.3\%$),

open-ocean ($18.2 \pm 11.6\%$), and deep-sea ($15.3 \pm 10.1\%$) networks. This pattern indicates that termination propensity is source-dependent and not strictly ordered by a marine-versus-river split. End-product fraction is the share of CRAM nodes lacking outgoing edges. It is not hydrolytic turnover, not carboxyl-driven recalcitrance, and not residence time.^{34,41,37}

This pattern does not require the docking panel to show hydrolytic shielding by carboxyl density. Outgoing CRAM-to-CRAM mass-difference edges include hydroxylation and carboxylation classes; that connectivity is not observed cycling. Docking reports pose energies of the same network ligands, not interception of carboxylated substrates in the environment. These are complementary rather than identical axes: network topology records mass-difference connectivity, whereas docking scores are in silico binding energies, not catalytic turnover.^{35,42,43}

Together, these results show that annotation-supported CRAM networks are environmentally structured: NPClassifier mixes give alkaloid-associated labels a major share of CRAM nodes alongside terpenoids and shikimates (Supplementary Table S3), while formation edges deplete alkaloid-like and enrich terpenoid-like labels; reaction-class balance remains oxidation-rich everywhere with riverine alkylation elevated (Supplementary Table S2) and alkylation-to-hydrolysis emphasized in riverine Markov structure. Annotation-consistency indices remain source-heterogeneous (Table 2; Supplementary Table S8), providing a quantitative topological readout of how CRAM-like connectivity is maintained across settings. Bacterial docking of the same ligands (Section 4) further shows that these structures are compatible with diverse bacterial enzymes, completing a structure cascade enabled by the expanded library.^{5,9,35,41,42}

Section 5: Bacterial enzyme docking of network CRAM ligands

To test whether network CRAM ligands are sterically and electrostatically compatible with bacterial enzymes that process carboxylated or aliphatic substrates, we docked annotation-supported network structures with smina (AutoDock Vina scoring function; exhaustiveness = 16, nine poses) against six crystal structures: a carboxylesterase (PDB 1EVQ; *Pseudomonas fluorescens*), β -ketoadipate decarboxylase (PDB 3AHD; *Acinetobacter baylyi*), an alkane monooxygenase (PDB 5T3Q; *Geobacillus thermodenitrificans*), a ring-cleaving dioxygenase receptor (PDB 2P4K), a marine bacterial esterase (PDB 3W9U), and a fungal laccase as a control (PDB 1GYC; *Trametes versicolor*). Receptors were prepared at pH 5, 7 and 9 and scored across environmentally relevant temperatures (2–32 °C). The 1,324 unique ligands and 846 unique A→B transformation pairs from 98 source files, yielding 84,235 individual poses and 53,854 competitive A/B records (Supplementary Fig S.15; Table 3). Docking results are shown in Figure 5, with additional poses and energy distributions in Supplementary Fig. S15.

Enzyme identity dominated the energy distribution. Competitive mean $E(A+B)$ was strongest for alkane monooxygenase (-8.06 ± 0.98 kcal mol⁻¹; $n = 8,988$ records) and weakest for β -

ketoadipate decarboxylase (-5.33 ± 0.77 ; $n = 8,964$), with marine bacterial esterase (-7.38 ± 1.03),
 laccase (-6.98 ± 0.82), ring-cleaving dioxygenase (-6.51 ± 1.29) and carboxylesterase (-6.40 ± 0.73)
 intermediate (Table 3). Individual-pose means reproduced the same ranking (global mean $\Delta G = -6.45$
 ± 1.34 kcal mol⁻¹; $n = 84,235$). Among chemotypes assigned in the campaign (benzene, alicyclic,
 aliphatic), benzene ligands bound most tightly (-6.84 ± 1.36) and aliphatic ligands most weakly (-5.72
 ± 1.11); alicyclic structures were intermediate (-6.56 ± 1.24) (Supplementary Table S10). Carboxyl
 count was weakly associated with stronger (more negative) ΔG (Spearman $\rho = -0.191$): mean ΔG was
 -6.20 at $n\text{COOH} = 1$ versus -6.86 at $n\text{COOH} = 3$ (Table 3). This direction is the opposite of a simple
 “more carboxyls, weaker hydrolase binding” model. Environment-averaged energies differed by <0.3
 kcal mol⁻¹ (river -6.53 to coral reef -6.27).

Reaction-pair $\Delta\Delta G$ values were modest relative to the enzyme-to-enzyme spread (mean
 competitive $\Delta\Delta G$ -0.06 to -0.28 kcal mol⁻¹ across the six enzymes; inferential $N = 1,324$ unique
 ligands / 846 unique pairs; SDs 0.67 – 1.08). pH 5 versus 7 versus 9 produced only small shifts
 ($|\text{Cohen's } d| \leq 0.10$ where significant; Supplementary Fig. S15). Docking therefore establishes that
 bacterial enzymes readily accommodate network CRAM poses, with enzyme identity—not carboxyl
 count—setting the dominant energy scale, and provides a computational bound on any simple
 carboxyl-driven hydrolytic-shielding increment at the pose-score level. Co-crystal redocking
 recovered rank-1 heavy-atom RMSD ≤ 2 Å (0.37 – 1.43 Å; Supplementary Fig. S15). 1EVQ was
 redocked against active-site HEPES rather than a native ester. These scores are computational pose
 energies, not measured k_{cat}/KM , and they do not by themselves explain marine CRAM residence times
 (Table 3; Supplementary Note 9).

Section 6 : Environmental implications

The central implication is that property-conditioned generative expansion of CRAM reference space
 measurably strengthens environmental MS/MS annotation and unlocks structure-constrained pathway
 analysis that formula-level libraries alone cannot support. Across 98 sources, marine CRAM chemistry
 is oxidation-rich with environmentally coherent network topology, while riverine networks emphasize
 a comparatively stronger alkylation-to-hydrolysis contribution; oxidation self-transitions remain
 prominent in non-CRAM internal and degradation subnetworks (typically 0.33 – 0.41 and 0.28 – 0.45),
 and formation subnetworks are more variable (0.20 – 0.40). Oxidation-to-carboxylation remains a
 persistent minority Markov branch (≈ 0.014 – 0.033), consistent with oxidation-led remodeling that
 diversifies CRAM space without requiring carboxylation as a single kinetic bottleneck. Source-level
 heterogeneity within each environment further indicates that local context modulates pathway
 signatures—an insight made accessible by the expanded library.

Limitations and next steps. Annotation-consistency indices and mass-difference edges are
 structure-supported topological hypotheses rather than radiocarbon residence times or enzyme-

validated reactions; isotope labelling and broader seasonal/depth sampling will further test the inferred sequences. Source counts remain imbalanced (notably coral reef $n = 4$), NPClassifier labels are out-of-distribution similarity statistics (65.26% high-confidence on pathway-assigned nodes), and 55% of the 476,647-spectrum corpus remains unexplained—room that the present library already begins to close. Native MetFrag isolate recovery (2/18 exact InChIKeys) and docking pose energies define clear frontiers for experimental follow-up (Supplementary Notes 1–10; Supplementary Fig. S15). These bounds do not diminish the principal advance: a CRAM-conditioned generative library that expands coverage, improves EPR, and resolves environmentally divergent annotation-supported pathways at campaign scale.

Methods

Overall study design

We developed a four-stage workflow: (i) CRAM-focused training set assembly and generative model training; (ii) generation and physicochemical screening of candidates; (iii) integration with environmental LCMS/MS data for annotation in the CRAM compositional region; and (iv) pathway analysis comprising reaction-network reconstruction, biosynthetic classification, and retention and sequential-reaction quantification. The objective was to test whether an expanded CRAM-specific candidate space changes annotation coverage and mass-difference network composition among 98 public LC–Orbitrap sources. Formation, origin, and residence time are not measured.

CRAM-focused training set and molecular representation

A curated training set of 5,842 CRAM and CRAM-like molecules was assembled to capture structural features relevant to carboxyl-rich alicyclic molecules (Figure 1b). The dataset combined 4,234 carboxylate-containing molecules from PubChem BioAssay, 928 alicyclic structures from natural-product collections (COCONUT and NPDB), 664 dicarboxylic acids from synthetic polymer precursor libraries, and 16 hypothetical CRAM-like structures derived from marine DOM spectroscopic reconstructions. Molecules were standardized using RDKit by removing duplicates, neutralizing where appropriate, and canonicalizing tautomeric forms. Canonical SMILES were used for indexing and comparison, and SELFIES strings were used for sequence decoding.^{17,3118,194445}

The training set was intentionally broad: molecules satisfying one or more CRAM-relevant characteristics (high carboxylate content, alicyclic character, or polycarboxylic composition) were included alongside strict CRAM end-members, exposing the model to candidate precursors and intermediates. Strict CRAM compliance (at least two carboxyl groups, $\geq 70\%$ aliphatic carbon, O/C ratio 0.4–0.6, at least one non-aromatic alicyclic ring) was evaluated separately during assessment.

Molecules were represented as atom-bond graphs; full atom feature definitions are provided in the Supporting Information.

Property-conditioned generative model

We trained a Property Conditioned Transformer VAE (Deep CRAM VAE backbone): Enhanced Hierarchical GAT (four layers, four heads, $H = 256$) fused with a Ring Chain Encoder, a 64-dimensional variational latent space, a 17D property head $\hat{y} = \text{MLP}_{\text{prop}}(z)$, and a six-layer causal Transformer SELFIES decoder (FF 2048, eight heads, dropout 0.1; 13.116 M parameters; validation loss 0.7381). Generation is not classical CVAE concatenation of properties into the decoder. Vocabulary size is 53; Length max = 150. TwoNN $d_{\text{int}} = 2.51$ and 9 PCs for 95% variance latent vectors.^{45,46}

Latent z is supervised by a 17D property head (size/polarity/rings/oxygenation/Hertkorn criteria) with MSE and decoded autoregressively by the Transformer SELFIES decoder. Full architectural specifications and the capacity sweep are in the Supporting Information.

Model training, molecule generation, and CRAM evaluation

The curated training set comprises $n = 5,842$ molecules (Figure 1b; 16 strict-CRAM exemplars), divided 85%/15% for training/validation. The objective is $L_{\text{recon}} + \beta L_{\text{KL}} + 0.1 L_{\text{contr}} + 0.5 L_{\text{prop}}$ with cyclic $\beta \in [5 \times 10^{-5}, 3 \times 10^{-3}]$ (four cycles), AdamW, cosine LR, AMP, and gradient clipping. The production checkpoint is the lowest-validation-loss model after 100 epochs (Property Conditioned Transformer VAE / ff2048). Enrichment R and Table 1 A use the curated denominator 16/5,842. Hyperparameter tuning, ablation analysis, and baseline model comparisons are described in Supplementary Note 2 and Supplementary Figs. S3, S4.

For generation, model performance was first evaluated on a representative 10,000-molecule validation sample to confirm that the model produced CRAM-enriched structures before proceeding to full-scale production. Full-scale production generation sampled 10 million latent vectors from the prior, conditioned on target property values, and decoded to SELFIES strings. Molecules were retained only if they passed RDKit sanitization and valence checks. Generated structures were then evaluated for strict CRAM compliance using the four criteria defined above. Structural diversity and novelty analyses (Morgan fingerprints, Tanimoto similarity, Murcko scaffold decomposition, scaffold entropy) and latent-space organization are reported in the Supporting Information.

Environmental plausibility screening

To prioritize candidate structures that are chemically realistic in aquatic environments, generated molecules were further screened using predicted physicochemical properties. Aqueous solubility was estimated using the ESOL model implemented in RDKit (Delaney 2004; RDKit: Open-

Source Cheminformatics Software 2023), and lipophilicity at seawater-relevant pH was assessed using predicted distribution coefficients at pH 8.1 (logD8.1). Molecules were considered environmentally implausible if predicted solubility fell below 330 $\mu\text{mol/L}$ (approximately 0.1 mg/mL for structures in the typical CRAM molecular weight range of 250-350 Da) or if logD8.1 exceeded 8. The 330 $\mu\text{mol L}^{-1}$ floor is a pharmaceutical solubility convention, not a seawater concentration cutoff. ESOL was benchmarked against 41 published aqueous-solubility values for CRAM-relevant reference compounds (RMSE = 0.71 log units, Pearson $r=0.83$); a post-hoc bias correction of +0.22 log units was applied before threshold evaluation. The $n = 41$ ESOL reference-compound list, and whether the +0.22 log-unit offset was estimated on those 41 compounds or a held-out subset, are not recovered from the packet. Sensitivity analyses at solubility cutoffs of 100, 330, and 1000 $\mu\text{mol/L}$ and logD8.1 cutoffs of 6–8 shifted the retained-candidate count by $\pm 11.4\%$ but did not alter the environmental ordering of downstream retention metrics (Supplementary Note 3 and Supplementary Fig. S6). All downstream analyses used the environmentally plausible subset; distributions are provided in the Supporting Information.^{3,8,34–36,47}

Environmental LCMS/MS data and candidate annotation

To evaluate the environmental relevance of the expanded CRAM library, we analyzed LCMS/MS datasets spanning coastal ($n=45$), deep-sea ($n=11$), open-ocean ($n=27$), coral-reef ($n=4$), and riverine ($n=11$) DOM, for a total of 98 samples. Open ocean is the largest spectral corpus in this campaign; the limiting source counts are coral-reef ($n=4$) and river/deep-sea ($n=11$), which remain recognized sampling limitations of this study; no seasonal, depth-stratified, or temporal-replicate sampling was performed, and ecosystem-scale claims are framed accordingly throughout. This study is entirely computational and in silico: no new wet-laboratory experiments, solubility measurements, incubations, or sample collections were conducted. All environmental LCMS/MS spectra analyzed here were obtained from publicly deposited datasets on MassIVE (accessions listed under Data availability). Sample collection, solid-phase extraction, and LC-Orbitrap acquisition for those datasets were performed by the original depositors under their published protocols, which are summarized in Supplementary Note 4 and Supplementary Fig. S9. Molecular formulas were assigned to the high-resolution MS peaks using conservative elemental constraints (C, H, O, N, S, and P), and formulas in the CRAM-like region of van Krevelen space were extracted for candidate annotation.

For each formula, candidate structures were queried from three reference scenarios: PubChem alone, PubChem supplemented with the generated CRAM library, and additional control libraries including COCONUT and ECMDB. MS/MS spectra were matched to candidate structures using fragmentation-tree-based scoring implemented through SIRIUS/CSI:FingerID (EPR in Results). The synthetic-isolate test used MetFrag (Supplementary Note 10) and is a separate engine; MetFrag is not a validation of SIRIUS annotations. Formula assignment and MS/MS annotation used SIRIUS 5 with

CSI:FingerID 5. MS¹ mass-error was tested at 1, 3, 5, and 10 ppm; in each test the MS² fragment-matching tolerance was set to the same ppm as that MS¹ setting. Headline Abstract/Results EPR and coverage statistics are the campaign annotation at 5 ppm (MS¹ = MS² = 5 ppm): TRFU n = 18,513 PubChem-only 31.8%, max-over-union 38.0%, CRAM-only 40.4%. Adducts were hydrogen-based and polarity-matched to each MassIVE dataset: [M+H]⁺ for positive-mode sources and [M-H]⁻ for negative-mode sources. Elemental constraints remained C, H, O, N, S, and P. A 5 ppm tolerance in reaction-network construction is a mass-difference edge cutoff, distinct from the SIRIUS MS¹/MS² ppm grid. For each scenario, explained peak ratio, candidate count, and top-candidate structural characteristics were recorded to quantify how expanded coverage altered annotation depth across environments.^{48,49}

For Synthetic-standard MetFrag filter-sensitivity test, we additionally tested whether class-prior chemotype sanitization of a formula-capped MetFrag changes injected paper-structure rank. Eighteen published standards (36 spectra, NCE 35 and 75) were compiled from literatures.^{3,34-36} Three paired protocols were run on the same spectra: (i) fair (generated CRAM only); (ii) inject (generated CRAM plus a PAPER overlay of the authentic SMILES, used only to measure true rank); and (iii) sanitization with injection was pre-filtered by class-prior chemotype rules before rematch. Filters used class priors only: exclude peroxide, carbonate, and mixed-anhydride SMARTS; require nCOOH ≥ 2; exclude aromatic rings except class aromatic with ether; and retain the van Krevelen envelope O/C 0.35–0.65, H/C 1.1–1.7, DBE/C 0.2–0.55. Candidate tables were formula-capped at the first 400 rows per formula. Because MetFrag Score=1 ties shared identical Explanation Peaks on 18/36 spectra and the paper structure was never in the Score=1 pool (0/36), offline Score=1 filters cannot raise paper rank; the complete Score=1 pool was retained.

Reaction-network reconstruction

To reconstruct molecular transformation pathways, we built directed mass-difference networks for each of the 98 environmental samples. Candidate edges were assigned when the exact mass difference between two annotated structures matched one of 27 predefined biochemical reaction classes within a 5 ppm tolerance. Directionality was assigned from the sign of the mass change relative to the reference transformation. Candidate transformations were then filtered using RXNMapper, which estimates atom-mapping plausibility for proposed reactions. Only edges with RXNMapper confidence scores of at least 0.5 were retained for downstream analysis. RXNMapper scores reflect atom-mapping plausibility, not thermodynamic feasibility. To partially address this, group-contribution ΔG° estimates (eQuilibrator; pH 8.1, 25°C, ionic strength 0.7 M) were computed for all 27 reaction classes; Markov-chain probabilities are reported both unweighted and thermodynamically re-weighted (Supplementary Note 6 and Supplementary Fig. S13).^{39,40}

Nodes were classified as CRAM-like or non-CRAM-like from the best-scoring annotation, enabling decomposition into non-CRAM→CRAM formation, CRAM→non-CRAM degradation, and internal cycling edges; all pathway interpretations are annotation-supported rather than structurally confirmed.

Biosynthetic pathway classification

All candidate structures across the 98 networks were classified using NPClassifier, which assigns hierarchical biosynthetic pathway labels on the basis of structural fingerprints, using the best-scoring candidate annotation per node. Pathway distributions were computed for precursor, intermediate, and end-product stages to track how biosynthetic composition changed during non-CRAM→CRAM transitions; cross-environmental comparisons used χ^2 -based tests. See Supplementary Note 7 and Supplementary Fig. S16.³⁰

Quantum-chemical reactivity profiling

Global molecular reactivity analysis combined conceptual-DFT (cDFT) descriptors with atom-resolved reactivity scoring across the generated CRAM candidate set. Electronic descriptors were computed with PySCF at the B3LYP-D3 level (6-31G(d) basis; Grimme's empirical dispersion correction D3) to obtain chemically interpretable global quantities (μ , η , softness, χ , and ω) and local response features (electrophilic/nucleophilic site indices and dual-descriptor extremes).⁵⁰ We summarized reactivity with molecule-level mean reactivity and atom-hotspot intensity rather than a raw maximum-task probability metric, because strict maxima are saturated by construction and are less informative for cross-molecule comparison. See Supplementary Note 9 and Supplementary Fig. S14.

Global descriptors followed standard cDFT finite-difference definitions from frontier-orbital energies, where ionization potential and electron affinity were used to compute $\mu = -(I + A)/2$, $\eta = (I - A)/2$, $\chi = -\mu$, and $\omega = \mu^2/(2\eta)$. Local susceptibility was represented by atom-condensed electrophilic and nucleophilic tendency scores and dual-descriptor extrema to distinguish broad molecular softness from site-selective activation. To improve comparability across diverse molecular sizes, hotspot summaries were evaluated jointly with heavy-atom count and molecular-weight covariates, and decile analyses were performed on rank-normalized reactivity scores rather than absolute maxima.

Descriptor reactivity relationships were evaluated using Spearman correlations between mean reactivity and cDFT/electronic descriptors (μ , η , softness, χ , ω), local dual-descriptor extremes, fingerprint densities, and size metrics (molecular weight, heavy-atom count). To characterize compositional shifts along the reactivity gradient, we compared top (90th percentile) and bottom (10th percentile) reactivity deciles for O/C, H/C, DBE, DBE/C, and elemental counts (C and

O). These analyses were used to connect quantum-chemical reactivity tendencies to the oxidation-dominated CRAM transformation signatures observed in the network layer.

Retention and sequential reaction analysis

To quantify persistence within the reconstructed networks, CRAM retention rate was defined as the fraction of outgoing CRAM edges that connected to another CRAM node rather than to a non-CRAM node. This metric was used to compare how strongly CRAM-like structures remained within CRAM transformation space across environments. End-product fractions were calculated as the proportion of CRAM nodes lacking outgoing edges. These measures are network-topological indices (annotation-consistency and end-product fraction), not persistence and not residence time. See Supplementary Note 6, and Supplementary Fig. S13.

First-order Markov chains were constructed from the directed networks; transition probabilities were estimated by normalizing RXNMapper-confidence-weighted counts across formation, degradation, and internal-cycling subnetworks to identify recurrent sequences such as oxidation-to-carboxylation and decarboxylation-to-ring-opening transitions.

Statistical analysis and reproducibility

Comparisons of annotation performance, reaction distributions, and environmental differences used non-parametric or categorical tests as appropriate, including Mann-Whitney U tests, Kruskal-Wallis tests with post hoc Dunn comparisons, and χ^2 tests. Summary values are reported as means $\pm$ standard deviations unless otherwise noted. All threshold choices, implementation details, baseline comparisons, feature definitions, supplementary pathway catalogs, and extended sensitivity analyses are in Supplementary Note 8 and Supplementary Fig. S17.

Computational docking protocol

Molecular docking was performed with smina using the AutoDock Vina scoring function (exhaustiveness = 16, nine output poses) against six enzyme crystal structures: carboxylesterase (PDB 1EVQ; EC 3.1.1.1, *Pseudomonas fluorescens*), β -ketoacid decarboxylase (PDB 3AHD; *Acinetobacter baylyi*), alkane monooxygenase (PDB 5T3Q; *Geobacillus thermodenitrificans*), ring-cleaving dioxygenase camphor receptor (PDB 2P4K), marine bacterial esterase (PDB 3W9U), and laccase (PDB 1GYC; EC 1.10.3.2, *Trametes versicolor*) as a control. Receptors were protonated at pH 5, 7 and 9; ligands were the annotation-supported nodes of source-resolved CRAM networks. Competitive A/B scoring used network-connected pairs (53,854 records after removing incomplete energies). The lowest-energy pose per ligand-condition record is reported as ΔG (kcal mol⁻¹). Co-crystal redocking used smina (exhaustiveness = 16, seed = 42) with symmetry-corrected heavy-atom RMSD; rank-1 RMSD ≤ 2 Å (Supplementary Fig. S15). See Figure 5 Supplementary Fig. S15.^{51,52}

Data availability

All data, code, trained model weights, and processed outputs supporting this study are publicly available at https://github.com/Suppboat2544/Generative_CRAM. The repository contains: (i) the curated 5,842-molecule CRAM-focused training set (canonical SMILES with property labels); (ii) the full trained Property Conditioned Transformer VAE / ff2048 model weights, architecture definitions, training scripts, and hyperparameter configurations; (iii) the generated environmentally plausible CRAM library of 5.04 million structures (CRAM.db) together with the physicochemical screening outputs; (iv) the 476,647 environmental MS/MS spectral indices and MassIVE accession (MSV000082086, MSV000082633, MSV000083098, MSV000082312, MSV000094541, MSV000094540, MSV000088937, MSV000083365, MSV000082950, MSV000092520, MSV000096004, MSV000095697, MSV000092622, MSV000093148, MSV000093514, MSV000092677, MSV000099559, MSV000089190 and MSV000089591), corresponding to the 98-sources. (v) bacterial docking export (Supplementary Note 9); (vi) 98 sample-level reaction-network edge tables with RXNMapper confidence scores, NPClassifier pathway assignments, and thermodynamic re-weighting coefficients; (vii) explainability outputs (GAT attention matrices, SHAP attribution tables, latent-traversal trajectories). The stated deposit does not regenerate Figures 1–5 and Tables 1–5 in full. Source MS/MS data are additionally archived on MassIVE under the accessions listed above.

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818 **Tables**

819 **Table 1. Study overview. Environment-resolved corpus detail and matching rates are in Table 2 and Supplementary Tables S5 and S7**

Metric	Value
Training set composition	
Total curated molecules / strict-CRAM exemplars	5,842 / 16
PubChem BioAssay carboxylates	4,234
Natural-product alicyclics (COCONUT/NPDB)	928
Synthetic polymer-precursor dicarboxylic acids	664
Hypothetical CRAM-like reconstructions	16
Train/validation split	85% / 15%
Model (Property Conditioned TransformerVAE / ff2048)	
Encoder	Hierarchical GAT (4 layers, 4 heads, H = 256) + RingChainEncoder
Latent / property head / decoder	z in $\mathbb{R}^{64}$ / 17D MLP _{prop} (z) / 6-layer Transformer SELFIES
Parameters / validation loss	13.116 M / 0.7381
mu Iticolumn2110k audit sample	
Valid SMILES / parseable	97.2% (9,721) / 96.8% (9,683)
Strict CRAM (all four criteria)	57.6% (5,760); 59.3% of valid
Environmentally plausible	70.0% (7,000)
Enrichment R vs training occupancy (16/5,842)	210.3 [128.9, 343.2]
Monte Carlo 95% CI (independent binomial)	[139.1, 377.3]

Metric	Value
MolGAN baseline (strict CRAM; no property head)	27.6%
KS vs marine DOM (O/C, H/C, NOSC)	D = 0.08, 0.11, 0.09
Full-scale library (10 M latent)	
Valid SMILES / CRAM-compliant / env.-plausible	9.72 M (97.2%) / 7.23 M (72.3%) / 5.04 M (50.4%)
Env.-plausible as % of CRAM-compliant	69.7%
Unique Murcko scaffolds / scaffold entropy	3.95 M / 21.8 bits (99.5% of theoretical max)
Environmental MS/MS annotation	
LC-MS/MS corpus	476,647 spectra / 98 sources
Spectral hits: PubChem / CRAM / COCONUT / ECMDB	155,994 / 64,572 / 27,428 / 4,487
CRAM-region formulas (PubChem -> +CRAM)	4,296 -> 7,069
Isomer candidates / mean per formula	5,401 -> 13,519 / 1.48 -> 1.64
EPR, TRFU CRAM features (n = 18,513; 5 ppm)	
PubChem top-hit -> max-over-union	31.8% -> 38.0% (+6.2 pp)
CRAM-library hits alone	40.4%

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825 **Table 2. Environment-resolved annotation and formation fingerprints. Full database counts, transformation enrichment, NPClassifier mixes,**
826 **Markov transitions, and RXNMapper scores are in Supplementary Tables S2–S5 and S9–S12.**

Env.	n	Spectra	PubChem	CRAM	Match %	O/R	Ox %	Alk %	Cons %
Coastal	45	106051	30508	13364	34.8±11.3	2.76	28.4±8.9	12.8±6.2	43.9±15.8
Deep Sea	11	42291	17177	7571	46.0±11.9	3.23	35.2±7.6	11.2±3.3	49.3±13.5
Coral Reef	4	11527	5206	1776	37.3±6.1	3.63	34.1±5.8	11.9±3.7	34.5±23.7
Open Ocean	27	269259	84015	35576	37.9±17.3	3.24	32.4±15.0	14.6±8.8	42.8±23.7
River	11	47519	19088	6285	33.4±13.2	2.75	29.4±9.9	15.3±8.1	44.7±29.4
Total	98	476647	155994	64572	—	—	—	—	—

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828 Match % = source-file matching rate (mean; Supplementary Table S7; library compatibility, not aging). O/R = oxidation/reduction formation means; Ox % /
829 Alk % = oxidation and alkylation shares of formation edges (Supplementary Table S2). Cons % = CRAM annotation-consistency mean (Supplementary Table
830 S8 ± SD; not residence time). End-product %: Coastal 22.4, Deep Sea 15.3, Coral Reef 27.6, Open Ocean 18.2, River 20.7. Network density highest in river
831 (0.0415; Supplementary Table S1).

832 **Table 3. Bacterial enzyme docking of network CRAM ligands (1,324 unique ligands / 846 pairs; 84,235 poses / 53,854 competitive records).**
833 **Chemotype/nCOOH detail in Supplementary Table S10; extended enzyme metrics in Supplementary Table S6.**

Enzyme / stratum	N	Mean E(A+B) or DeltaG	Median E	Mean DeltaDeltaG	Coop. %
Competitive pairs by enzyme					
Carboxylesterase	8892	-6.40 ± 0.73	-6.3	-0.08 ± 1.08	52.9
beta-Ketoadipate Decarboxylase	8964	-5.33 ± 0.77	-5.3	-0.14 ± 0.67	55.0
Alkane Monooxygenase	8988	-8.06 ± 0.98	-8.0	-0.11 ± 0.90	53.5

Ring-Cleaving Dioxygenase	8988	-6.51 ± 1.29	-6.6	-0.28 ± 0.96	57.9
Marine Bacterial Esterase	8802	-7.38 ± 1.03	-7.3	-0.08 ± 0.96	51.2
Laccase (control)	8988	-6.98 ± 0.82	-7.0	-0.06 ± 0.73	52.8
Individual poses by chemotype					
benzene	35995	-6.84 ± 1.36	—	—	—
alicyclic	24814	-6.56 ± 1.24	—	—	—
aliphatic	23426	-5.72 ± 1.11	—	—	—
Individual poses by nCOOH					
nCOOH = 1	43461	-6.20 ± 1.43	—	—	—
nCOOH = 2	32775	-6.68 ± 1.19	—	—	—
nCOOH = 3	7193	-6.86 ± 1.18	—	—	—
nCOOH = 4	806	-6.61 ± 1.21	—	—	—

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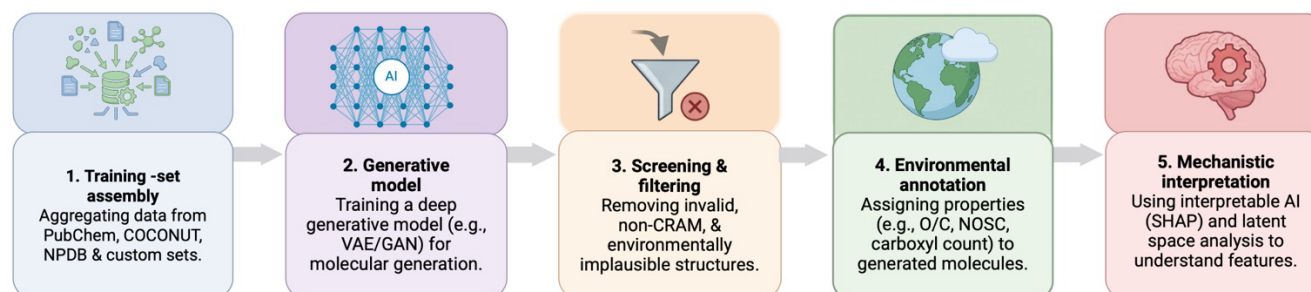
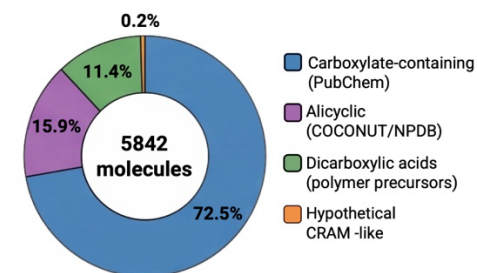
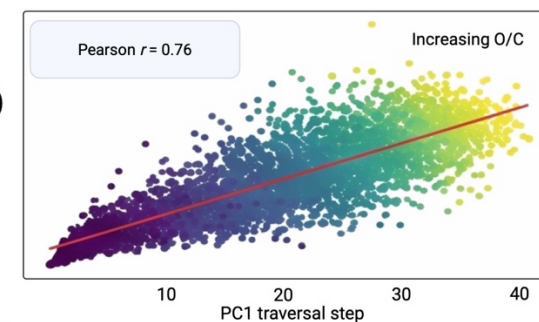
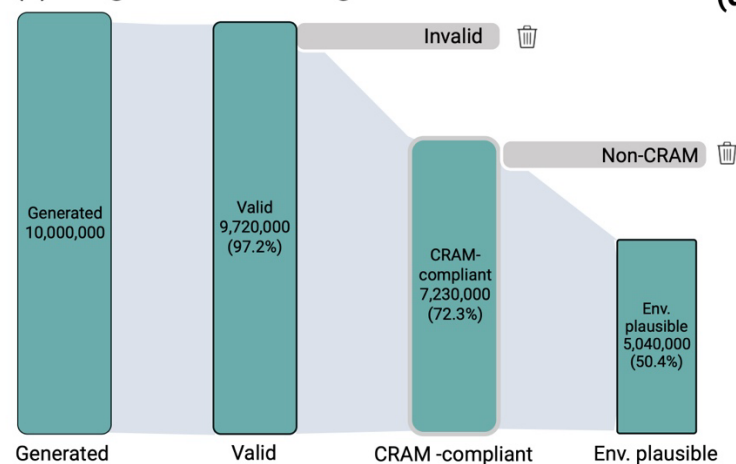
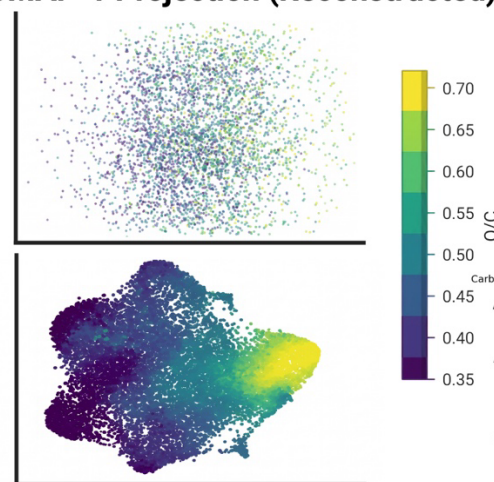
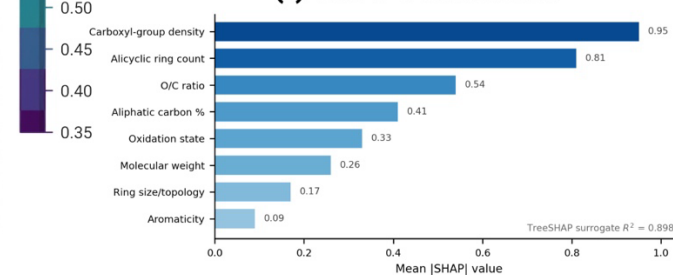
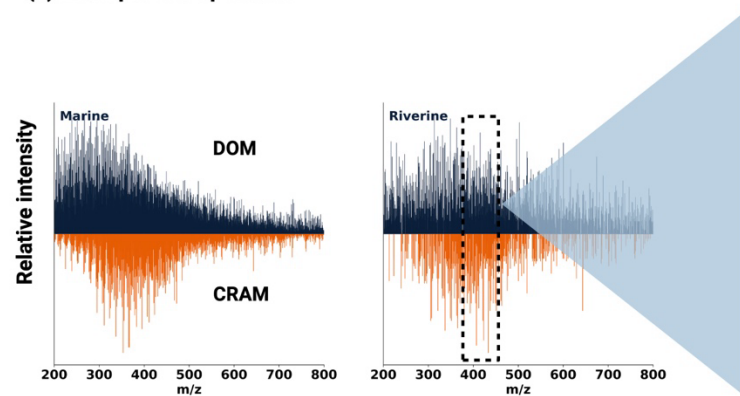
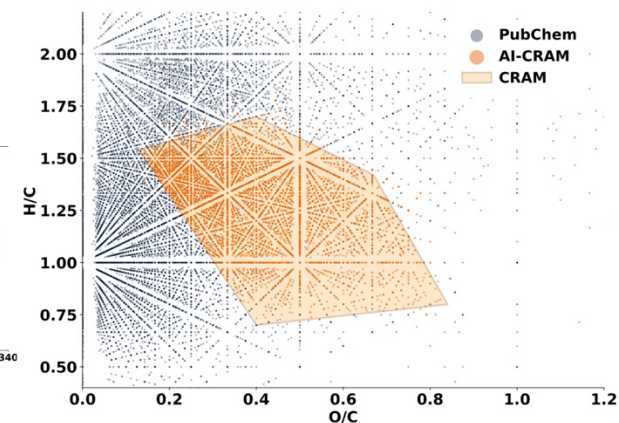
(a) Overview of study**(b) Training Dataset Composition****(e) PC1 Latent Traversal****(c) Progressive Filtering CRAM Database****(d) UMAP-1 Projection (Reconstructed)****(f) SHAP Attribution**

Figure 1 AI -assisted reconstruction of environmentally relevant CRAM structural space: (a) Schematic overview of the AI -assisted workflow, from CRAM training-set assembly and molecular generation to environmental annotation and mechanistic interpretation; (b) Composition of the 5,842-molecule training dataset and its major structural classes; (c) Diagram showing progressive filtering of generated molecules from 10 million candidates to 5.04 million environmentally plausible CRAM structures (Full database); (d) UMAP projection of the learned 64-dimensional latent space colored by O/C ratio, illustrating organization along an oxidation gradient; (e) Latent-space traversal along the first principal component (PC1) showing continuous increases in O/C ratio across generated molecules; and (f) SHAP feature importance identifying carboxyl-group density and alicyclic ring count as the dominant determinants of CRAM prediction.

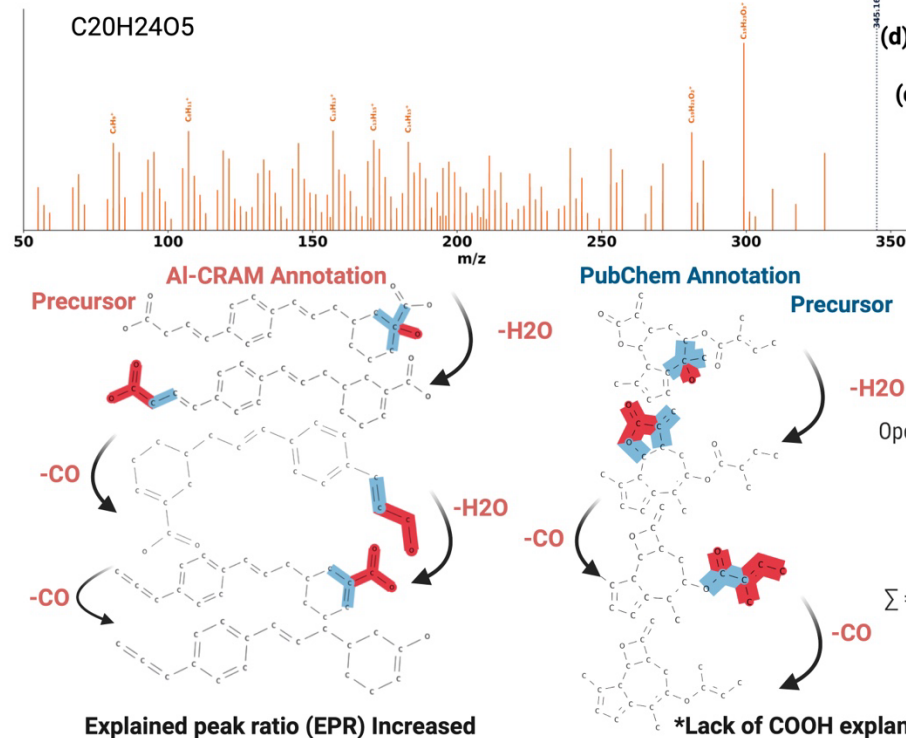
(a) Example MS Spectrum



(b) Van Krevelen Diagram (CRAM Compositional Space)

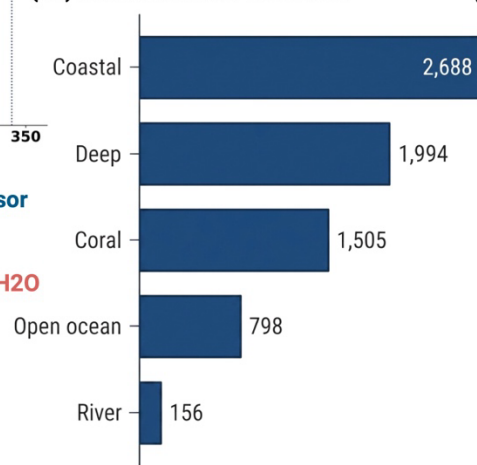


(c) Representative LC-MS/MS Spectrum with Fragmentation Tree

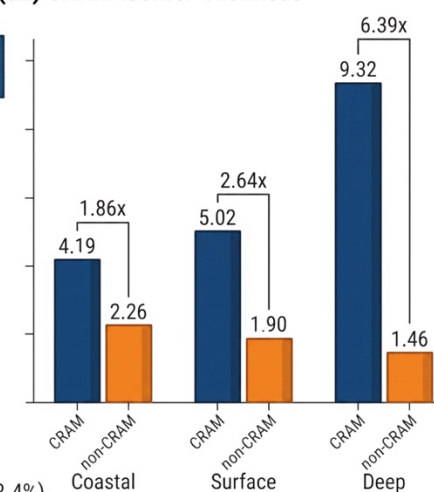


(d) Expansion of CRAM Formula and Isomer Coverage

(d1) CRAM Formula Richness



(d2) CRAM Isomer Richness



$\Sigma = 7,141$ · distinct from 1,364 unique formulas (18.4%)

Figure 2. AI-CRAM structural library integration improves CRAM-region annotation coverage and spectral explanation of marine dissolved organic matter (DOM) mass spectra. (a) Example MS spectrum. Representative broadband mass spectrum (m/z 0–800) with CRAM-region features highlighted in red against other DOM features in gray. Inset (m/z 380–430) resolves individual CRAM-region peaks (e.g., m/z 401.1294, 425.1547), illustrating the fine mass resolution required to distinguish carboxyl-rich alicyclic candidates from the broader DOM feature envelope. (b) Van Krevelen diagram of CRAM compositional space. O/C (x-axis, 0–1.2) vs. H/C (y-axis, 0.4–2.2) scatter of PubChem-matched formulas (teal) and formulas added by AI-CRAM (orange). The CRAM region is delineated by CRAM pentagon. Right inset: a verified MS/MS fragmentation example anchoring the diagram — the CRAM-region parent ion, $C_7H_5O_4^-$, m/z 153.0193) loses CO_2 (43.9898 Da) to give the deprotonated fragment ($C_6H_5O_2^-$, m/z 109.0295). (c) Representative LC-MS/MS spectrum with fragmentation tree annotation. Side-by-side comparison of the same spectrum annotated using PubChem only (left, blue; explained peak ratio, EPR = 31.8%) versus PubChem + AI-CRAM combined (right, red; EPR = 38.0%). The AI-CRAM-supplemented annotation resolves substantially more of the m/z 350–800 fragment envelope, corresponding to a +6.2 percentage point gain in EPR ($n = 18,513$ spectra evaluated). (d) Expansion of CRAM formula and isomer coverage. (d1) AI-CRAM-exclusive annotation occurrences by environment (Coastal 2,688; Deep 1,994; Coral 1,505; Open ocean 798; River 156; $\Sigma = 7,141$), distinct from the 1,364 unique AI-CRAM-only formulas (18.4% of 7,393 total annotated formulas) with no matching PubChem structure — exclusivity is concentrated in oceanic rather than riverine samples. (d2) Isomer richness by environment, comparing CRAM-classified vs. non-CRAM isomers per formula: Coastal 4.19 vs. 2.26 (1.86 $\times$ enrichment), Surface 5.02 vs. 1.90 (2.64 $\times$), Deep 9.32 vs. 1.46 (6.38–6.39 $\times$), based on 7,500 total formulas of which 5,142 are CRAM-classified. Deep-water samples show the largest isomer-hypothesis expansion under AI-CRAM integration.

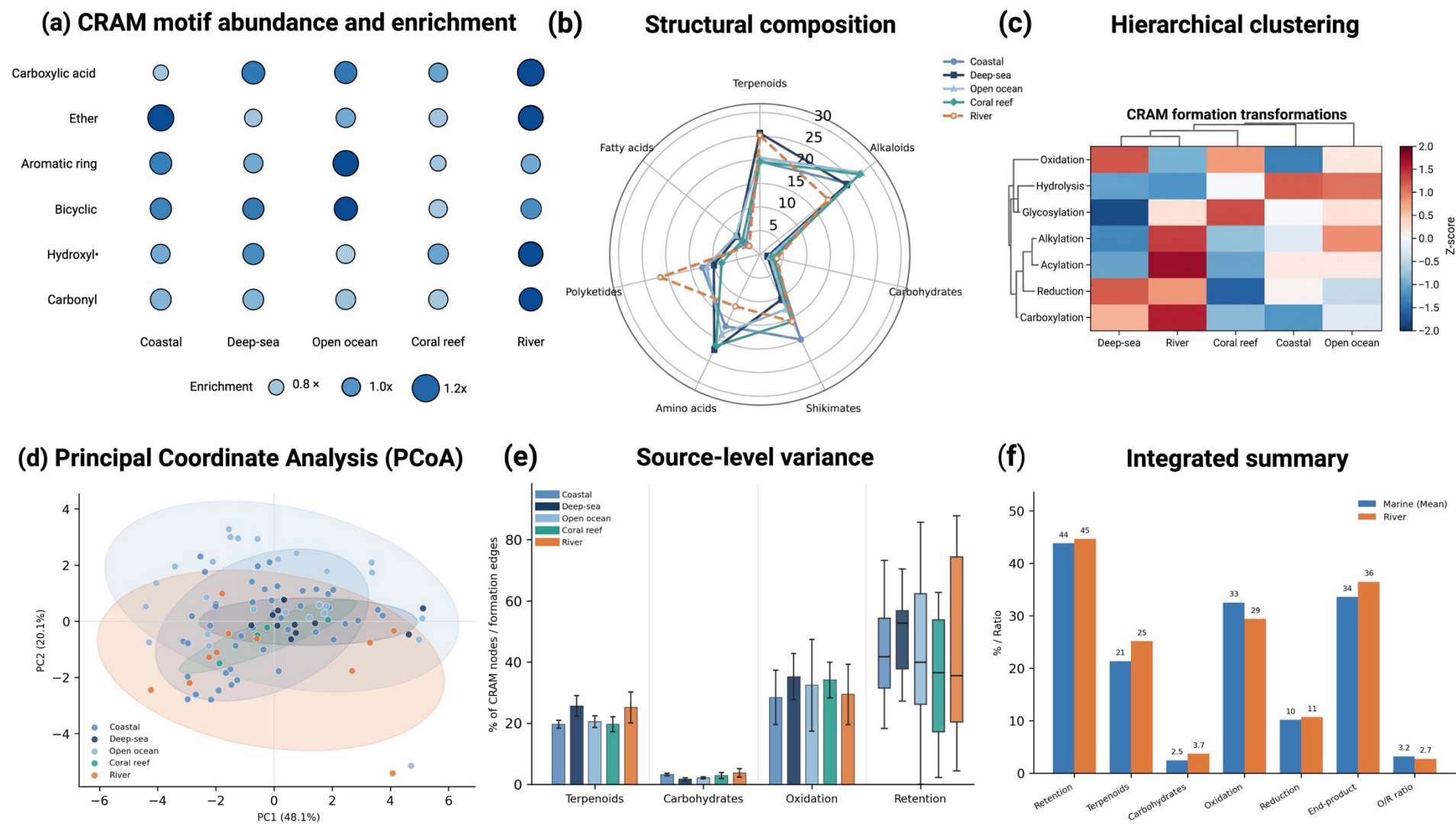
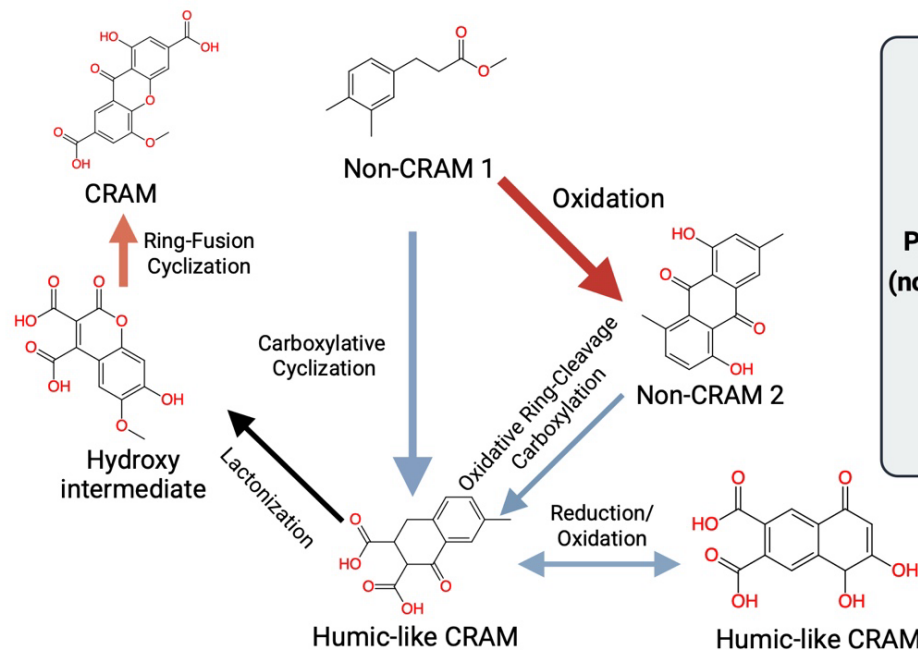
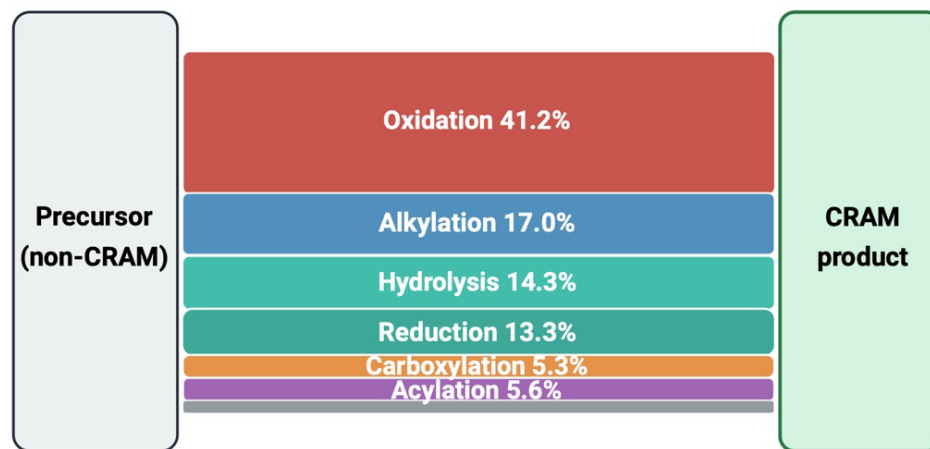


Figure 3 Structural motif abundance, compositional variance, and formation-pathway clustering of CRAM across coastal, deep-sea, open ocean, coral reef, and river dissolved organic matter (a) CRAM motif abundance and enrichment — carboxylic acid, ether, aromatic ring, bicyclic, hydroxyl, and carbonyl motif enrichment (0.8×–1.2×) across the five environments, with river showing the highest carboxylic acid and hydroxyl enrichment. (b) Structural composition — radar plot of precursor-class contributions (terpenoids, alkaloids, carbohydrates, shikimates, amino acids, polyketides, fatty acids), showing terpenoids and alkaloids dominate across all environments, with river deviating most from the marine environments. (c) Hierarchical clustering of CRAM formation transformations — z-scored heatmap of oxidation, hydrolysis, glycosylation, alkylation, acylation, reduction, and carboxylation contributions, clustering deep-sea/coral reef/coastal separately from river. (d) Principal coordinate analysis (PCoA) — overall compositional ordination (PC1 48.1%, PC2 20.1%) showing river as the most distinct cluster, with marine environments overlapping substantially. (e) Source-level variance — box/bar plot of terpenoid, carbohydrate, oxidation, and retention percentages per environment, showing retention has the widest variance of the four metrics. (f) Integrated summary — marine-mean vs. river comparison across retention, terpenoids, carbohydrates, oxidation, reduction, end-product, and O/R ratio metrics.

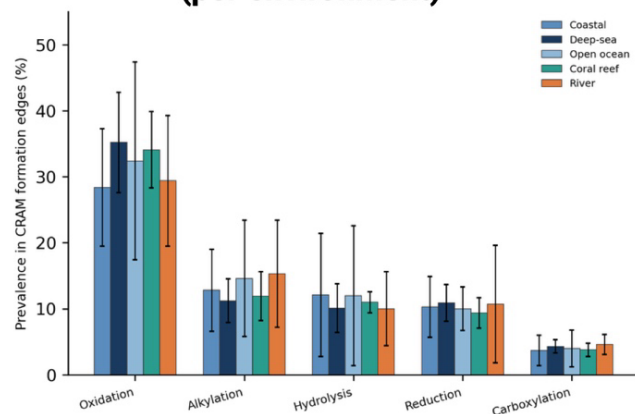
(a) Molecular transformation network (schematic)



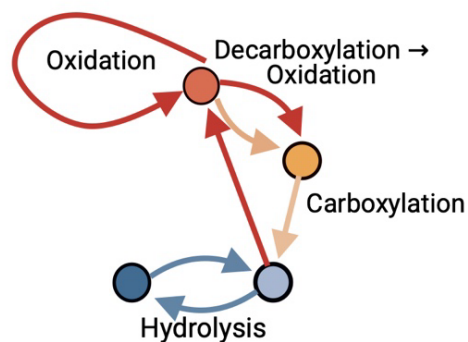
(b) CRAM formation



(c) Reaction-class abundance (per environment)



(d) Directed Markov transition network (sequential)



(e) Conceptual oxidation-carboxylation flow model

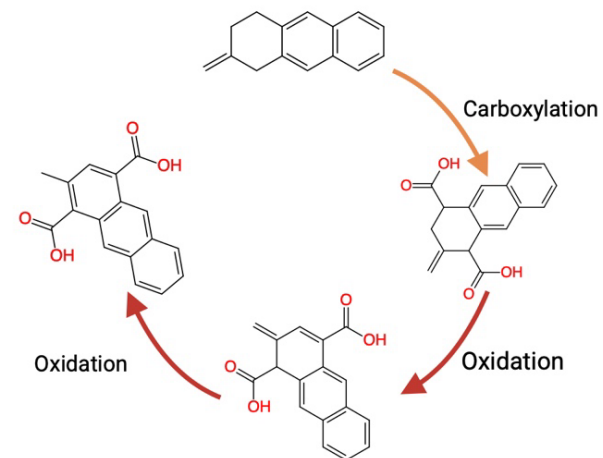


Figure 4. Molecular transformation network and reaction-class dynamics underlying CRAM formation from non-CRAM precursors (a) Molecular transformation network (schematic). Traces the full structural route from a simple non-CRAM ester precursor to mature CRAM products, with each arrow now labeled by its correct named reaction: Oxidation (Non-CRAM 1 → Non-CRAM 2), Oxidative Ring-Cleavage Carboxylation (Non-CRAM 2 → Humic-like CRAM), Carboxylative Cyclization (Non-CRAM 1 → Humic-like CRAM), Reduction/Oxidation (the reversible redox couple between the two Humic-like CRAM structures), Lactonization (Humic-like CRAM → Hydroxy intermediate), and Ring-Fusion Cyclization (Hydroxy intermediate → CRAM). (b) CRAM formation. A funnel/flow diagram quantifying the relative contribution of each reaction class across the whole precursor-to-CRAM-product transition: Oxidation dominates at 41.2%, followed by Alkylation (17.0%), Hydrolysis (14.3%), Reduction (13.3%), Acylation (5.6%), and Carboxylation (5.3%). (c) Reaction-class abundance (per environment). Bar chart with error bars showing how prevalent each reaction class is within CRAM-formation edges across the five environments (coastal, deep-sea, open ocean, coral reef, river) — oxidation is consistently the most prevalent class everywhere, with broadly similar patterns across environments. (d) Directed Markov transition network (sequential). Models which reaction classes tend to follow which in sequence — showing, for example, that hydrolysis commonly precedes carboxylation, and that decarboxylation frequently couples directly into a subsequent oxidation step. (e) Conceptual oxidation–carboxylation flow model. Illustrates a representative molecular example of the dominant carboxylation-then-oxidation sequence on a real ring system, showing how sequential carboxylation and oxidation events on an anthracene-type scaffold progressively build up a polycarboxylic, more oxidized CRAM-type product.

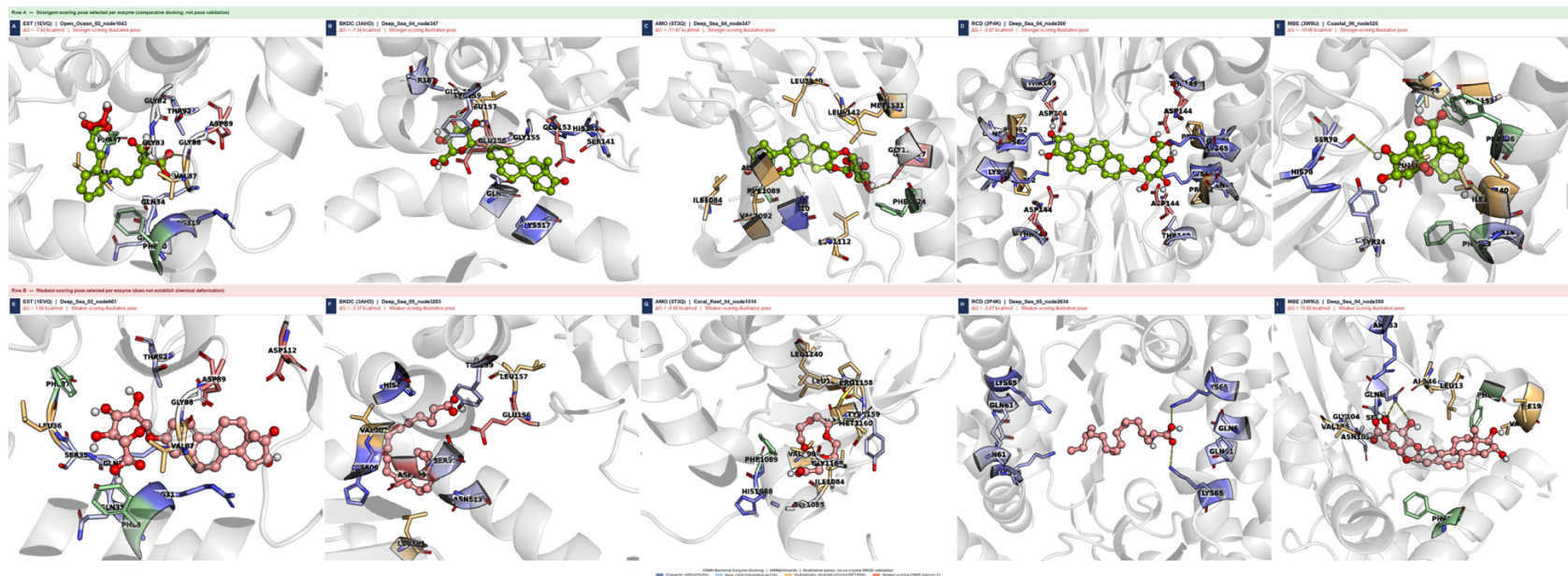


Figure 5. Illustrative comparative docking of CRAM-type ligands against candidate biotransformation enzymes: strongest- and weakest-scoring poses. Docking poses for five candidate enzymes are shown for comparative, illustrative purposes only; poses are not validated against crystallographic reference ligands and do not establish chemical degradation. (Row A, green header) Strongest-scoring pose selected per enzyme from comparative docking. (Row B, pink header) Weakest-scoring pose selected per enzyme, shown for contrast; a weak score does not establish, and should not be interpreted as evidence against, chemical degradation. Columns correspond to: EST (PDB 1YEG), DHDC (PDB 3AHD), AMO (PDB 5TSQ), RCO (PDB 2F4X), and MSE (PDB 3W9J). Each panel lists the source CRAM node identifier (e.g., Deep_Sea_04_node347, Coastal_06_node525) and the corresponding docking score (ΔG , kcal/mol). CRAM ligand carbons are colored green (strongest-scoring pose, Row A) or salmon/white (weakest-scoring pose, Row B); protein carbons are colored by enzyme; key active-site residues are labeled and shown in stick representation. Docking was performed using SMINA/Vinardo scoring on a CRAM structural backbone in a comparative screening mode; poses are illustrative only and were not subjected to RMSD validation against experimentally resolved ligand geometries.